Discovering Plate Approach Patterns Among Major League Baseball Batters Through Data Mining

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***Abstract*—This project examines patterns in Major League Baseball (MLB) batting behavior by analyzing plate approach statistics from qualified batters over the 2021-2025 seasons, including swing, contact, walk, and strikeout rates. The analysis uses the Baseball Savant MLB leaderboard, a publicly available dataset, and focuses on discovering natural sets of plate approaches from player-season observations based on the data in the leaderboard. After preprocessing and standardizing the data, as well as engineering and selecting the appropriate features, K-Means clustering was applied to identify natural groupings of hitters based on their approach at the plate. Three distinct clusters emerged, representing selective, swing-heavy, and contact-oriented batters, each with unique characteristics and tendencies. Selective batters showed the highest overall offensive value and expected offensive value, while other batters generated similar results to each other. To extend beyond clustering, anomaly detection techniques were implemented to identify players who either deviate structurally from their assigned cluster or significantly over- or underperform relative to expected outcomes, measured by the difference between weighted on-base average (wOBA) and expected weighted on-base average (xwOBA). This found both extreme adherence to and hybridization between plate approaches, as well as volatility in performance, most influenced by contact, but mostly randomized due to external factors. Overall, the analysis demonstrates that batter plate approaches exist on a spectrum, and that combining clustering with anomaly detection provides deeper insight into player profiling, emphasizing strategy alongside observed results. These findings contribute to the understanding of offensive behavior in baseball, allowing for a more accurate evaluation of players.**

I. Introduction

The purpose of this project is to discover patterns in Major League Baseball (MLB) batter data, revealing the diverse strategies players utilize at the plate. This data highlights the underlying discipline and traits a player holds: Are they an aggressive or passive hitter? Do they make consistent contact when they swing? These types of metrics determine where each player excels, describing their skill and style regardless of their end-line success, which is useful for evaluating and comparing them on a comprehensive level.

Furthermore, this understanding bridges the gap between plate approach and performance. Traditional baseball analysis often focuses on results like batting average, rather than underlying statistics, ignoring the various ways hitters achieve those outcomes. Many plate approaches exist across the league, and players may achieve similar results using very different plate approaches, or vice versa, something that goes unnoticed if examining only performance.

The objective, therefore, is to group players into distinctive archetypes for analysis, uncovering the natural points where hitting styles diverge. These specific insights will reveal plate identities and allow for a deeper understanding of players, invaluable primarily for MLB organizations when making recruitment or tactical decisions. Additionally, locating profile anomalies, both in plate approach and performance, will find players who deviate from dominant trends, extracting valuable exceptions to patterns.

II. Dataset Description

The large dataset chosen for this project comes from a publicly accessible source and captures several critical features of the data it represents. There are also drawbacks to the dataset that must be addressed.

A. Data Source

The dataset used is derived from the Baseball Savant MLB leaderboard, and the data was exported through the platform's built-in CSV download functionality. The dataset displays player behavior and performance data among qualified MLB batters over the past five seasons (2021-2025), defined as players with at least 500 plate appearances in a single season.

B. Data Size

In the selected dataset, there are 669 rows, each representing a player-season observation, and 11 columns representing plate approach measurements and outcome statistics.

C. Data Representation

The dataset describes individual plate discipline and tendencies, as well as performance quality for an overall season. Plate approach statistics track how often a batter swings at a pitch, where they swing, and how frequently they make contact with the ball. These measurements relate to results, summarized by weighted on-base average (wOBA), signifying the observed outcomes of each player's performance. Expected weighted on-base average (xwOBA) is included to measure underlying performance based on the quality of contact made, rather than actual results, excluding outcome variance due to luck, and putting more of the focus on plate approach. The data only

accounts for a player's overall performance across an entire season, not a game-by-game analysis.

D. Key Measurements

Observation Data

- Player
- Season

Plate Approach Data

- K% - Strikeout Rate
- BB% - Walk Rate
- Z-Swing% - In-Zone Swing Rate
- O-Swing% - Out-of-Zone Swing Rate
- Z-Contact% - In-Zone Contact Rate
- O-Contact% - Out-of-Zone Contact Rate
- Swing% - Overall Swing Rate

Performance Data

- wOBA - Weighted On-Base Average
- xwOBA - Expected Weighted On-Base Average

E. Sample Data Table

A miniature sample data table is given below in Table 1 to provide a visualization of how the data is formatted. This sample shows a condensed form of the dataset across three observations and six attributes.

F. Data Quality Considerations

There are limited data quality issues in the dataset. Firstly, the dataset is restricted to qualified batters, excluding part-time or bench players, which favors the analysis toward established, full-time players. As a result, it is likely that more high-performing players will appear than low-performing ones, as players who perform very poorly are unlikely to receive full playing time across a whole season. Additionally, because all data is taken at a season level, any in-season variation in individual plate approach or performance is masked, hiding any possible dramatic shifts in those areas. The dataset also does not account for the opposing pitcher, varying in-game situational plate appearances, or stadium differences, which may influence plate approach and outcome statistics.

III. Discovery Questions

The following discovery questions will guide the objectives of the project and determine the mining techniques implemented. Their combined value will ensure the soundness of the overall analysis and its conclusions.

TABLE I
SAMPLE DATA

Player	Year	K%	BB%	wOBA	xwOBA
Tucker, Kyle	2021	15.9	9.3	.383	.394
Olson, Matt	2025	24.3	12.6	.366	.360
Swanson, Dansby	2024	24.3	9.1	.307	.324

A. Questions

1) Are there distinct, natural groupings of MLB batters based on their plate approach statistics? : This is the key question that will be answered: finding the patterns arising from how batters approach the plate and the groupings that form from them.

2) How do observed offensive results compare to expected results across these groupings? : This question focuses on the offensive results of the plate approach profiles, finding if there is a group that performs better or worse among them, and also if a group over- or underperforms significantly.

3) Are there players whose plate approach or performance makes them extreme outliers compared to their own typical profile or the rest of the league? : This question will drive the identification of observations that feature offensive approaches and performances that do not align with the usual results seen from their respective plate approach or overall player set.

B. Value

These discovery questions are valuable because they reveal broad plate approach profiles among MLB batters, which will be evaluated against each other in different ways. Performance outcome and variance comparisons will identify the strongest performing player archetypes and any patterns that emerge through them. Finding outliers enhances the analysis by uncovering pattern exceptions, which offer insight into which plate approaches differ significantly or tend to result in over- or underperformance when compared to expectations. All of these goals contribute to the understanding of plate approaches throughout MLB and support accurate analysis of players and their performances.

IV. Implemented Data Mining Techniques

The core data mining of this project will involve two techniques: clustering and anomaly detection. Clustering will form the foundation, profiling players based on their plate approach, while anomaly detection will extend this analysis by identifying outliers who deviate significantly from their established profile, or in terms of their offensive performance.

A. Clustering

K-Means Clustering will be employed to determine whether distinct, natural groupings of MLB batters exist based on their plate approach statistics. Unsupervised learning directly addresses this goal, finding genuine patterns in the data without the influence of external elements. Only the swing rates, contact rates, and other discipline measures of players are considered. The method also verifies the validity of spherical data types and ensures that the patterns found within them are concrete.

B. Anomaly Detection

Anomaly detection will be primarily performed using distance-based techniques to achieve two goals.

Firstly, Euclidean distances from each data point to its cluster centroid will be calculated to identify players who

drift furthest from their assigned cluster. These are edge-case hitters who do not align exactly with one plate approach but instead adopt a hybrid strategy, or batters who showcase extreme instances of their plate approach and favor towards one-dimensionality.

Then, anomaly detection will be used to identify player-seasons that greatly over- or underperformed their expected outcomes compared to all other batters across all plate approach profile groupings. Examining differences between wOBA and xwOBA statistics through z-scores within clusters will directly locate this unusual behavior. This may indicate disproportionate skill or variance in a batter, or just an unlikely occurrence of a plate approach profile deviating from typical patterns, providing insight into whether certain plate approaches are more likely to cause excessive over- or underperformance.

The first anomaly detection implementation focuses on extreme and integrated plate approaches, while the second concentrates on irregular performance divergence.

V. Analysis Pipeline

Fig. 1 depicts the data analysis pipeline of the project. Data inspection and preprocessing will consist of collecting, cleaning, and finalizing the dataset. Exploratory data analysis will be conducted using detailed univariate, bivariate, and multivariate analyses. Feature engineering, selection, and transformation will finalize the features to be used in clustering and ensure the data is transformed in a way that properly facilitates clustering algorithms. Finally, the data mining stage will apply the techniques described in Section IV to the data to highlight its patterns and characteristics, whilst also drawing interpretative value from them.

VI. Exploratory Data Analysis

Exploratory data analysis began by cleaning and confirming the validity of the dataset. This was followed by examining each of its features, individually and collectively, to understand their distributions and relationships.

A. Data Inspection & Preprocessing

The preparatory inspection and preprocessing of the dataset found no critical errors. Each row and column held valid values and were properly filled in, and no duplicate values were encountered in the data. The preliminary distributions of each feature were normal and standard in terms of central tendency and overall dispersion.

Fig. 1. Data Pipeline Flowchart

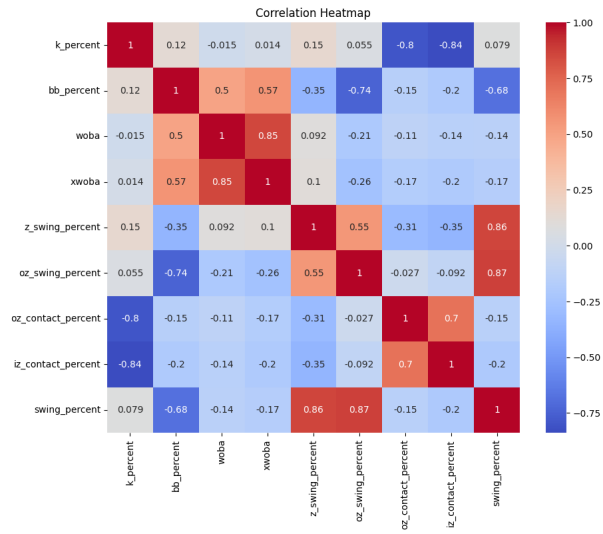

Fig. 2. Correlation Heatmap of Features

B. Univariate Analysis

The univariate analysis of each feature yielded results similar to one another. Meaningful variation in swing behavior, contact ability, and discipline was found in the attributes, and their distributions exhibited a unimodal distribution on their respective scales. Variations included right and left skews, revealing several outliers at both ends of some charts. In short, these examinations produced no prominent results, but they did provide useful context. This context paved the way for feature comparison, where combined relationship analysis uncovered the intricacies of the dataset as a whole.

C. Bivariate and Multivariate Analysis

When comparing these features through bivariate and multivariate analysis, direct and inverse relationships appeared. Strong positive correlations arose specifically between wOBA statistics and walk rate, while key negative relationships occurred between strikeout rate and contact percentages, as well as between walk rate and swing rates. The correlation heatmap represented in Fig. 2 revealed these patterns, and subsequent scatterplots were formed to confirm them. Together, they showed that the statistics grow in meaning when examined together, establishing the need for a diverse set of features to demonstrate pivotal relationships in the dataset.

VII. Feature Engineering & Selection

Feature engineering required the addition of a new characteristic measuring the performance difference in players. Additionally, feature selection narrowed down the attribute set to include only the core traits necessary for later data mining and analysis.

A. Feature Engineering

The dataset needed one feature modification before beginning the data mining phase. A performance difference statistic was created to measure the difference between a

player's wOBA and their xwOBA. This data point provides the essential information to view whether a player over- or underperformed in a given season and by how much, indicated by a positive or negative value, without taking into account the strength of their recorded performance outcome.

For example, a player with a wOBA and xwOBA of 0.245 and 0.235, respectively, shares an equivalent performance difference with a player who achieved a wOBA and xwOBA of 0.392 and 0.382, respectively. Performance difference values will be ideal for identifying clusters that stand out in this category to determine which plate approaches are stronger or weaker than observed due to varying amounts of variance.

B. Feature Selection

Most of the plate approach and performance data listed earlier were preserved. They all measure the critical decisions batters make at the plate, the approaches that form from those decisions, and their resulting expected and observed outcomes on the field. The only removal was the Swing% characteristic. As found in the bivariate analysis, this statistic served only as a mean between Z-Swing% and O-Swing%, rendering its purpose redundant. Therefore, it was omitted to clarify cluster creation and produce clearer relationship evaluations.

VIII. Data Transformation

The data was transformed for clustering through standardization. The features in the set existed on a percentage scale, but their ranges were not properly aligned with each other. For example, K% held a spread between 3%-34%, while Z-Swing% held one between 52%-83%. These conflicts would have biased distance-based algorithms like K-Means, and therefore had to be correctly coordinated. Standardization solved this problem by making each range proportional to the others for accurate comparisons. This decision retains variability and outlier handling while centering each feature around zero with consistent standard deviations, ensuring equal clustering contributions.

Fig. 3. Elbow Plot

Fig. 4. Silhouette Scores Plot

IX. Clustering

The clustering process consisted of selecting a suitable clustering algorithm, choosing the appropriate number of clusters, and implementing the technique. The formed clusters revealed insights regarding plate approaches and offensive results, providing solutions to the first two proposed discovery questions.

A. Algorithm Selection

The K-Means clustering algorithm was chosen for cluster creation because the dataset was found to be normally distributed with continuous and standardized features, making it well-suited to spherical cluster centers that represent precise and meaningful batter profiles.

B. Number of Clusters

The process of choosing the number of clusters to pull from the dataset was performed by using the elbow method and taking corresponding silhouette scores for k values. Based on the elbow method, represented by Fig. 3, a value of k = 3 was selected as the optimal value for maximizing in-cluster variance while minimizing the total number of clusters. The most competitive silhouette scores came from k values k = 2 and k = 3, illustrated in Fig. 4. A k value of k = 2 held a score of nearly 0.26, while a value of k = 3 maintained a score of just under 0.25, conveying a difference of approximately 0.01.

Despite k = 2 being the technically stronger option, a total of three clusters were decided to draw from the dataset. The elbow method result and minuscule silhouette score discrepancies support this because three groupings make for a richer and more diverse body, broadening relationship understanding. This decision retains knowledge without sacrificing dataset complexity or key attributes, avoiding oversimplification and resulting in a more comprehensive layout for in-depth analytical, evaluative, and interpretive purposes.

C. Cluster Characterization

Clustering implementation found three distinguishable clusters. A grouped bar chart, shown in Fig. 5, and later varied scatterplots indicated these sections of groupings and visualized them distinctly.

1) Cluster 1: Cluster 1 ('Selective Batters', 196 points) holds hitters who show patience at the plate. They have the lowest swing and contact rates but the highest walk and strikeout rates, displaying extreme plate discipline, for better or worse. Their performance is also significantly the best of the clusters, and they notably underperform their expected statistics as well. In theory, this makes the selective plate approach the best in terms of average offensive value.

2) Cluster 2: Cluster 2 ('Swing-Heavy Batters', 251 points) holds hitters who demonstrate lots of plate aggression. They hold the highest swing rates at the price of high strikeout and low walk rates. While some batters can perform very well when implementing this strategy, many can also suffer from it, defining the approach as a double-edged sword. Their performance falls much more in line with the mean wOBA, and they underperform that number by a very small amount.

3) Cluster 3: Cluster 3 ('Contact Batters', 222 points) holds hitters who make well-above-average contact with balls, holding the clear best numbers in this category. They also strike out the fewest between the three clusters. Evidently, their utility comes from putting the ball in play, opening up opportunities on the field, and leaving more room for luck than usual. Performance-wise, they are on par with Cluster 2 but slightly overperform their expected statistics.

D. Plate Approach Findings

The first proposed discovery question can now be answered. There are distinct groupings of MLB batters based on their plate approach: discipline-based, contact-based, and swing-heavy. The clusters all feature unique characteristics and lead in the approach measurements for their specific specialties, as characterized above. Each approach exhibits its own patterns and territory, cementing cluster clarity.

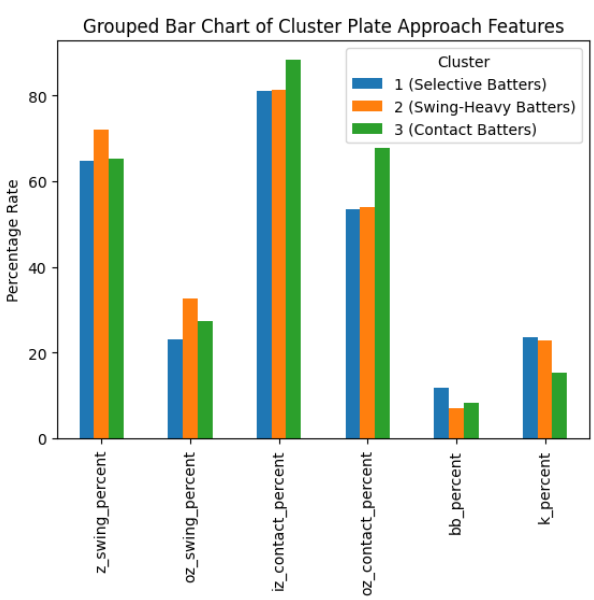

Fig. 5. Grouped Cluster Bar Chart

The selective batters of Cluster 1 are very sound and suggest a methodology of patience and temperament. They are picky and work the ball-strike count, increasing the chances of a walk or strikeout.

The swing-heavy batters that make up Cluster 2 are aggressive and attempt to take the maximum advantage of every pitch thrown. Gambling on this strategy is risky, as these players can end their at-bats with a home run or a strikeout on three pitches, which would be far from the ideal plate appearance.

Cluster 3 batters rely on their bat-to-ball ability rather than a clear-cut swing approach. The emphasis is that when they do swing, they make contact, putting the ball in play more often than striking out. This improves the luck factor of baseball and opens up opportunities along the basepaths for other players.

Batters with exceptional bat-to-ball talent tend to fall under Cluster 3, while the other two clusters are mainly defined through their swing rates: low swing rates in Cluster 1 and high swing rates in Cluster 2.

E. Offensive Results Findings

The second proposed discovery question can also be resolved. Observed and expected offensive results (wOBA and xwOBA) differentiate most within Cluster 1, which marks a considerable advantage over the other clusters' similar offensive metrics.

The offensive value of selective batters is generally the most effective and could be considered the most ideal. Expected result statistics are also most strongly pronounced in Cluster 1. This group largely underperforms relative to expectations, whereas the other clusters only slightly over- or underperform. This notable distinction provides further evidence as to how well selective batters perform against other clusters through prominent underlying measurements.

Cluster 1 batters hold the greatest quality of contact and subsequent results, allowing them to reach base more effectively than others.

F. Limitations

There are limitations to these takeaways. The performance of Cluster 1 could be influenced by batting order placement, situational at-bats, or the fact that this cluster holds the smallest sample size by a noteworthy margin. While further analysis is required to validate these determinations, the preliminary trends appear to hold. Most importantly, players can relate to any cluster and still perform well or poorly.

X. Structural Anomaly Detection

Structural anomalies are based purely on batter plate approaches within each cluster. This detection process involved taking Euclidean distances between data points and centroids to pinpoint the highest and lowest values. The results found

outlier players and hybrid players who held unique plate approaches relative to their designated profiles, expanding the scope of player evaluation beyond cluster assignments and revealing new discernments, structural patterns, and strategic implications.

A. Approach

To find deviations within the structures of each cluster, Euclidean distances were taken between each data point and its respective cluster centroid. The top 5% of distances for each cluster were identified as anomalies, a threshold chosen because it flags only statistically unusual points without being overly strict. Additionally, Euclidean distances were drawn to the other cluster centroids, allowing differences to be measured between the closest cluster and the next closest one. These differences were marked as hybrid scores, and the lowest 5% of these values were similarly gathered as anomalies. These two sets of anomalies were labeled as outlier anomalies and hybrid anomalies, where outliers represented players far from their cluster centroid, and hybrids represented players at the boundary between two clusters.

B. Results

Two sets of anomalies were identified: outliers and hybrids. Outlying players characterized intensely committed player plate approaches, while hybrid players were specified as batters not committed to a singular approach.

1) Outliers: Outliers were identified as instances of extreme plate approaches within each cluster. These batters essentially doubled down on the aspects of their plate approaches that set them apart within their respective cluster, exemplifying exaggerated, unusual profiles. The Fig. 6 scatterplot displays this by highlighting the edges of the graph where all of the outliers remain.

For example, these underscorings include players who hold significantly high contact rates outside the zone, indicating Cluster 3 hitters that possess overwhelming contact ability. There are also players with extremely low swing rates outside the zone present, indicating batters from Cluster 1 who are exceptionally disciplined.

Each cluster contained a nearly symmetrical distribution of outliers: 10 in Cluster 1, 13 in Cluster 2, and 12 in Cluster 3. This shows a balanced cluster structure with similar overall variance. In total, 35 outliers were flagged, which aligns with 5% of the dataset as previously intended. Large ranges across attributes existed within each cluster as well, indicating scattered spreads on all fronts.

2) Hybrids: Hybrid batters were identified as players holding a mixture of two plate approaches. Their split location between cluster centroids suggested this, depicted in Fig. 6 by the highlighted players creating the borders around the separated cluster areas.

For example, a player caught between Clusters 1 and 2 may be labeled as a swing-heavy batter, but has the plate discipline commonly found in selective batters in the form of a low O-Swing%. In such a case, the player must be assigned to one

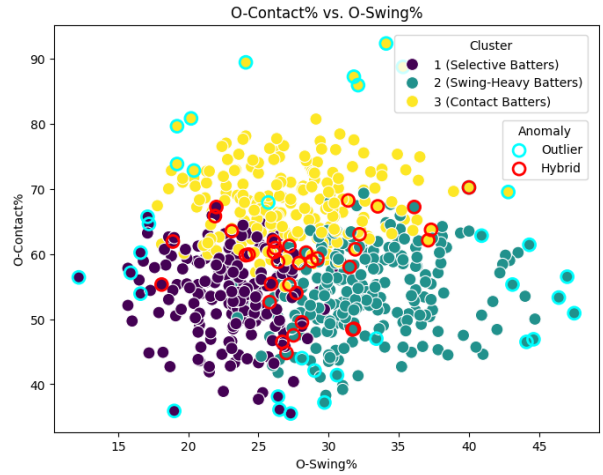

Fig. 6. Cluster Scatterplot with Structural Anomaly Highlighting

of the formed clusters, but in reality, hovers in the middle of them.

Additionally, the attribute ranges of hybrids were more tightly packed and similar, the opposite of what outliers produced. However, hybrid player cluster distributions were identical to those of outliers: 10 in Cluster 1, 13 in Cluster 2, and 12 in Cluster 3. This again shows a proportional cluster structure and variance, and 35 total hybrids flagged likewise lines up with 5% of the dataset.

C. Nature of Plate Approach Anomalies

Regarding diverging plate approaches, outlier and hybrid players emerged, each representing unique changes in their plate approaches relative to their cluster grouping.

Some players are so entrenched in a plate approach that they become outliers because of the extreme nature of their statistics. These outliers showcase the span of individual approaches and pinpoint exact points rather than subjectively inferring them from plots.

Conversely, hybrid players do not fit squarely within one cluster. They live in the overlaps, indicating that not every player fits perfectly under one umbrella in the way that K-Means clustering imposes. Because the clustering phase forced a single assignment, these batters were previously misrepresented despite having prominent features across multiple clusters.

D. Structural Patterns

Treating these anomalies as an extension of the clustering process reveals clarity that the clusters alone could not communicate. It exposes a flaw in the process: categorizing data points into single clusters when they could have qualified for another. Mapping these hybrid points discerns set-in-stone cluster points from borderline ones, allowing a deeper understanding of player strategies.

Anomalies appear to be natural, balanced phenomena occurring across the board. Because the cluster structure has similar overall variance, no single group is more likely to hold

structural anomalies than any other. They simply outline the two opposite sides of a single cluster: Some players fit almost too well into their group, while others barely fit because of their split between two groups.

E. Strategic Implications

Identifying outliers and hybrid anomalies is meaningful because it further clarifies a player's hitting type, going beyond the original clustering assignments.

Outlier players are more static and predictable because they are radically specialized in their approach. Exaggerated, over-optimized strategies lead to these unusual profiles because those players are more constructed as one-dimensional or specialized hitters with pronounced strengths and weaknesses. These outliers can also represent the boundaries of each cluster's identity, serving as reference points for the cluster's range.

Hybrid players, however, are more dynamic and adaptable, as they can utilize skills from different plate approaches in a versatile way. They display multidimensional skill sets, challenging the assumption that all players fit neatly into a single plate approach category. Therefore, clustering provides useful abstractions, but still holds gaps that these players exploit.

F. Limitations

The limitations of this anomaly detection may include misclassifying points as anomalies when they are, in reality, boundary points. Another issue is that these anomalies are based on model-defined clusters, and thus may not be fully accurate when looking at these data points without clustering applied.

XI. Performance Anomaly Detection

Performance anomalies are based on offensive results across all players. Detecting these anomalies involved taking z-scores of performance differences to find values farthest from the mean. The results found players who over- and underperformed, uncovering no patterns but revealing the role that contact has in favorable variance.

A. Approach

Major performance anomalies within each cluster were identified by standardizing each data point's performance distance metric using z-scores. This method found the most unusual cases of performance discrepancies based on differences between wOBA and xwOBA. Z-scores more than 2 standard deviations from the mean were defined as anomalies, with positive values indicating overperformance and negative values underperformance.

B. Results

34 total performance anomalies were located, heavily favoring overperformers. This total matches 5% of the dataset, aligning with the expected number of points not in the [-2, 2] standard deviation range of the mean. However, no clear patterns could be discerned between either category of performing players.

1) Overperformers: 20 overperforming players were found, with Cluster 3 accounting for half of them. The overall distribution is as follows: 4 in Cluster 2, 6 in Cluster 1, and 10 in Cluster 3. Clusters 1 and 3 showed the most players holding the largest performance difference values. Fig. 7 plots overperforming players, and while they show a large range in Z-Contact%, no clear pattern differentiates them from underperforming players aside from cluster denotements; overperformers appear most frequently in Cluster 3 in the high range of Z-Contact%.

2) Underperformers: 14 underperforming players were found, evenly spread throughout the clusters: 4 in Cluster 1 and 5 in Clusters 2 and 3. Cluster 2 was the most prominent group with the most extreme cases of underperformance. Similar to overperformers, no evident pattern appears when plotting underperformers. They display a tighter range, but remain mostly scattered on the graph in Fig. 7.

C. Nature of Performance Anomalies

Over- and underperformance is difficult to analyze because many factors drive the large disparities between wOBA and xwOBA. This gap represents the difference between what should happen based on how the ball is hit and what actually happens on the field. Because these outcomes are influenced by defensive alignment, stadium dimensions, and environmental conditions like wind, massive performance differences are naturally unpredictable. Consequently, there is no concrete way to predict these discrepancies or their specific causes; they underline luck and external context rather than plate approach alone.

D. Variance of Contact

While performance anomalies are volatile, the imbalance across clusters suggests a controllable factor is present. Frequent bat-to-ball contact opens more opportunities for favorable luck when the ball is in play, explaining why the overperformer distribution favors Cluster 3 contact batters. As the only controllable factor in the wOBA calculation, contact

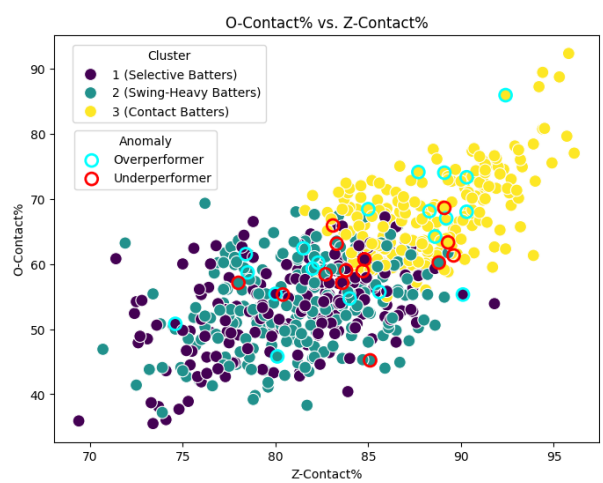

Fig. 7. Cluster Scatterplot with Performance Anomaly Highlighting

increases offensive potential; conversely, limited contact from Cluster 2 swing-heavy batters leads to the opposite, with only four overperformers in that group. Cluster 1 selective batters fall in the middle, generating more impressive numbers than Cluster 2 but fewer than Cluster 3.

This suggests that while contact is not the sole controller of wOBA, given the significant variance it exhibits, it is the one element a batter can affect at all times. Because there is higher outcome volatility tied to balls in play, Cluster 3 is more susceptible to favorable variance.

E. Performance Distributions

A key finding is the distinction between types of luck: extremely good luck can be encountered more easily through specific plate approaches, while bad luck remains largely constant across the league. Underperforming players were consistent across the data, with equal cluster distributions. These underperformers reflect unrealized potential and may signal future upward regression as opposed to overperformers who experience short-term success beyond expected metrics. This indicates that bad luck is fixed across the league and uncontrollable, while good luck can be slightly altered by a player's approach.

F. Limitations

Ultimately, performance anomalies remain mostly random and contribute to a significant limitation of this analysis. Players within the same cluster, even with very similar underlying statistics, can have wildly varying results. While these patterns can be identified, it is impossible to fully account for infinite external variables that finalize outcomes. This reinforces the reality that overall baseball performance is a mix of skill and unpredictable randomness that exists independently of a batter's plate approach.

XII. Structural & Performance Anomalies Combined

The two types of anomalies shared rare cases of intersection. Additionally, the last proposed discovery question can be answered from the complete anomaly detection process.

A. Anomaly Overlap

Very few of the anomalies coincided with each other, as seen in Fig. 8. This suggests a lack of connectivity between them, meaning that no anomalies make the presence of another anomaly more likely than before. Unnatural plate approaches will still mainly yield normal performance results, whereas normal plate approaches outlined by the clusters can randomly produce unexpected performance results.

B. Overall Anomaly Results

The final proposed discovery question can be answered using the results of the comprehensive anomaly detection. There are outlier player approaches and performances that appear across player clusters. Structural anomaly detection revealed deviating plate approaches, which included outliers and hybrids. Outliers were players who were highly committed to

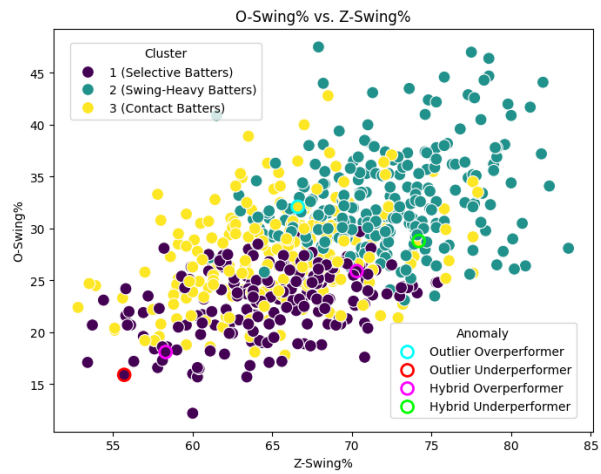

Fig. 8. Cluster Scatterplot with Combined Anomaly Highlighting

their distinguished plate approach, while hybrids were players who were undecided between two different approaches. Performance anomaly detection, on the other hand, indicated several instances of players overperforming and underperforming their expected offensive results across the data.

XIII. Critical Assessment

The methods instantiated in this project were thoroughly validated. Additionally, all limitations and potential ethical concerns were addressed to mitigate mishandlings and misunderstandings of the findings.

A. Validity

The stability of the findings from this analysis is supported by consistent results across repeated K-Means runs. The model ensures reproducibility by using a fixed random state, with repeated runs confirming that the batter profiles exhibit persistent structures rather than artifacts of a single initialization. While the selection of k = 3 was supported by the elbow method, the narrow margin between k = 3 and k = 2 silhouette scores suggests that alternative groupings may have also been possible.

However, these results should be viewed as exploratory insight rather than definitive conclusions. The clusters and anomaly sets are shaped by methodological choices, including feature engineering, transformation steps, and thresholds for anomaly detection. The preliminary patterns are logically supported, but remain sensitive to these parameter adjustments. Consequently, further refinements or additional data could potentially shift these boundaries, altering the perceived results.

B. Limitations

The process and results of this study are subject to several important limitations, summarized here.

First, the dataset is restricted to qualified MLB batters who totaled at least 500 plate appearances in a single season. This biases the findings toward full-time starters and excludes part-time and poor-performing players who do not receive

consistent playing time, limiting league-wide representation. Furthermore, external game factors such as stadium dimensions, weather conditions, and defensive positioning are not captured, despite frequently having a significant influence on plate approaches and performance outcomes.

Methodologically, K-Means clustering assumes spherical structures with equal variance, which may oversimplify the complex relationships between different batter profiles. Similarly, anomaly detection depended on distributional assumptions and specific threshold values, which could have misclassified borderline cases or overlooked subtler anomalies.

Finally, the scope of this project is centered on pattern discovery rather than prediction. As a result, the findings cannot definitely determine why certain approaches lead to certain outcomes or if they represent a persisting trend. While these results likely remain relevant for other full-length MLB seasons, they should not be inferred for different leagues under different conditions.

These limitations imply that the findings should be viewed as an initial framework for understanding plate approaches rather than absolute truths.

C. Ethical Considerations

The findings of this project carry low ethical risk, as the analysis utilizes publicly available MLB player statistics rather than sensitive or private data. However, concerns lie in the misuse of these analytical results if they are treated as conclusive judgments of player value. The prioritization of specific plate approaches believed to be optimal could lead to reductive evaluations of players in organizational scouting and decision-making.

To mitigate this, these results should be treated as contextual insights that add to player understanding rather than determinations of their potential. Real-world applications must incorporate surrounding information to ensure a fair and balanced evaluation of all players. This is crucial, as the limited amount of numerical statistics captured in this analysis make up only a fraction of the many facets that define a player's contribution and performance.

XIV. Conclusion

Through exploratory data analysis, feature transformation, clustering, and anomaly detection, this study has laid out useful patterns in individual MLB batter data that define modern plate approaches. Players naturally align with one of three distinct strategies: discipline-based, swing-based, and contact-based; each carrying varying performance outcomes. Discipline-based batters yielded the highest average offensive value, while swing and contact-based approaches produced similar results. The detection of anomalies further refined these archetypes, identifying outlier and hybrid players who lean toward opposite sides of clusters, challenging their boundaries. Additionally, the analysis of performance unusualities revealed that contact rates influence positive outcome volatility in overperforming players, while underperformance showed symmetry throughout, despite such variability remaining heavily dictated by external factors.

These findings provide a comprehensive outline of various batting archetypes across MLB, offering a deeper understanding of player tendencies beyond surface-level observed results. By emphasizing approach alongside performance, this data can be utilized across the league to minimize bias in player evaluation. Practical applications of this research include player acquisition decisions, roster management, and lineup construction, and even reach consumer-facing platforms such as fantasy baseball. This work demonstrates that the combination of plate strategy and statistical results is essential for accurate player assessment.

Future work provides significant opportunities to expand on these findings greatly. Data expansion offers the most valuable possible avenues to explore: Incorporating pitch type, batting order, and situational context information could reveal the finer subtleties of patterns and determine how different circumstances influence a player's established approach. Future iterations can also benefit from exploring methods more deeply. Alternative clustering algorithms, such as Hierarchical clustering, can be used to account for non-spherical relationships, for example. Anomaly detection can also be refined by adjusting rules and thresholds to yield more precise patterns. This would offer even greater clarity regarding borderline players existing between cluster formations.

References

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